Hengji Zhang

Quantitative Research & Investment Data Analytics | Market Data Due Diligence, PIT Validation, Factor Verification

bosprimigenious@bupt.edu.cn · github.com/bosprimigenious · bosprimigenious.github.io · Beijing, China

PROFILE

Quantitative research and investment-data specialist with hands-on experience across two sell-side and buy-side internships: alternative-data tracking at a top-tier brokerage research institute and investment-data verification inside a leading securities asset-management arm. Work centres on whether a number can be used at a given decision date - source licensing, point-in-time (PIT) availability, entity mapping validity, unit and frequency consistency, coverage denominators and missing-value semantics - then on whether a signal survives out-of-sample and after trading costs. Certified CFA Level 2 candidate; builds reproducible, fully traceable workbooks rather than narrative reports.

CORE COMPETENCIES

Market data due diligence	source licensing, coverage denominators, PIT vs snapshot, publication calendars, revision history
Entity & instrument mapping	share-code reuse, cross-exchange duplicates (000001.SH vs 000001.SZ), fund-vs-portfolio aggregation, SPV chains
Missing-value taxonomy	UNKNOWN vs NOT_OBSERVED vs NOT_PROVEN; prohibiting zero-fill and its effect on coverage
Quant factor validation	IC / RankIC, layered backtests, decay curves, parameter grids, out-of-sample and survivorship control
Trading-cost economics	turnover-to-cost conversion, impact estimates, gross vs net return, capacity limits
LLM benchmark quality	rubric authoring, adjudication, six mandatory validation checks, hallucination case library

PROFESSIONAL EXPERIENCE

CITIC Securities · Asset Management — Investment Data Verification Intern

Mar 2023 - Present · Beijing

- Verify the factual queries behind investment-research outputs: 1,200+ alternative and market indicators (freight rates, capacity utilisation, production schedules, tenders, satellite and passenger-flow series) checked under a four-step template - raise, reconcile, evidence, conclude.
- Run three-way source comparison across Wind, Bloomberg, iFinD and third-party alternative vendors; classify every discrepancy into five causes (definition, update timing, entity mapping, unit/precision, discontinuation) so recurring breaks are fixed at the definition level rather than re-checked each cycle.
- Authored 80+ methodology documents, each stating definition, formula, frequency, unit, publication calendar, change log, known defects and applicability boundary - used downstream as the acceptance basis for queries.
- Maintain a change-and-sample-adjustment ledger (date, before/after value, affected scope) and a discontinuation register recording the last valid publication date, preventing stale snapshots from being consumed as current data.

Shenwan Hongyuan Securities · Research Institute — Cyclical Industries Research Intern (Transport & Commodities)

Feb 2021 - Feb 2023 · Beijing

- Co-maintained high-frequency alternative databases - container freight (SCFI / CCFI / BFI families), blast-furnace and cement utilisation, order books and production schedules, tender award values, freight volume and passenger flow - supporting 240+ weekly notes and thematic data commentaries across 10+ sub-sectors.
- Built the indicator ledger: publishing entity, frequency, history start, release time and licensing constraint for every series, which later became the denominator source for coverage checks.
- Cross-validated 1,500+ alternative indicators, reconciling same-source definition variants (composite vs route-level, tax-inclusive vs exclusive, MoM vs YoY base) and flagging definition changes and sample-universe adjustments.
- Applied seasonal adjustment and base restoration to freight-rate and utilisation series, and separated initial prints, revised prints and forecast vintages in working papers so revised values were never presented as first releases.

Independent Expert Engagement · Financial Benchmark Quality — Part-time Expert Reviewer (LLM Evaluation)

Jun 2024 - Present · Remote

- Review 1,300+ model responses for market-data query benchmarks: rubric scoring, ground-truth cross-checks and 180+ cross-reviewer adjudications; contributed sample screening, criterion decomposition, rubric calibration and

dispute post-mortems.

- Enforced six mandatory checks: PIT availability by publication date; entity-mapping validity windows including cross-market duplicate codes; unit and frequency consistency (index points vs percent, TEU vs FEU, daily vs weekly); coverage numerator/denominator alignment; missing-value taxonomy instead of zero-fill; and feasibility of documented replacement sources.
- Maintain a hallucination case library of 260+ instances (snapshot passed as PIT, missing values zero-filled, wrong coverage denominator, uncitable source, silent definition swap), classified by error type, trigger condition, decision boundary and correct handling; issued 110+ root-cause reports feeding the annotation standard.
- Compiled replacement-source lists for 70+ discontinued, low-frequency or broken series, recording channel, update cadence, definitional gap versus the original indicator and a comparability verdict.

QUANTITATIVE RESEARCH & FACTOR VERIFICATION

WorldQuant BRAIN · Alpha Research (concurrent with asset-management internship)

Alpha research and factor validation · Jun 2023 - Present

- Construct cross-sectional and event-driven factors from price-volume, fundamental and alternative inputs (freight, utilisation, passenger flow, tenders); each factor carries a registry entry for input fields, frequency, unit, availability date and source permission.
- Evaluate single factors with IC / RankIC, layered backtests, long-short spread and turnover; retain decay curves across holding periods, and archive factor definition, universe and industry-neutralisation scheme together so results produced under different parameterisations are never pooled.
- Test robustness through out-of-sample windows, parameter grids and year-by-market-cap stability, guarding against reporting the best in-sample parameter set as the result.
- Convert turnover into commission and impact cost to report net-of-cost return and capacity, rather than gross spread.
- Control look-ahead and survivorship: align strictly to publication date, use point-in-time index constituents, and archive delisted or removed names separately.
- Gate factor admission on data acceptance: coverage, missing-value type and update frequency bound the usable sample and therefore IC reliability, so sources failing acceptance never enter factor computation.

SELECTED PROJECTS

TextGuard - Chinese AI-generated-text detection platform — Project lead (undergraduate innovation programme)

Nov 2025 - Jul 2026

- Designed and built a full-stack detection platform (Python / FastAPI inference, React / TypeScript front end) with risk visualisation, explanation panel, sample comparison and exportable detection reports.
- Authored TextGuard-Bench (five-axis evaluation suite) and the MGAB held-out attack protocol; quantified in-distribution versus OOD separation with a G3 metric instead of a single accuracy figure.

MingChaWanXiang - multimodal research-integrity monitoring platform — Engineering intern, Information Network Engineering Research Centre, MOE

Jan 2026 - Jun 2026

- Contributed multi-stage recall and similarity search (CLIP + Milvus) to detect image reuse, paper-mill output and AIGC artefacts across text, figures, formulas, tables and citation networks.
- Supported evidence-chain construction and report export; every finding is traceable to a source location.

Zhilian 3.0 - university LLM practice platform — Full-stack intern, Information Network Engineering Research Centre, MOE

Jul 2025 - Jan 2026

- Implemented list, filter, pagination and record-version views in Vue 3 + TypeScript; aligned field definitions, error codes and record versions with backend, with empty-result, no-permission and timeout states surfaced explicitly.

PUBLICATION

- Hengji Zhang et al., "TextGuard: a dual-path semantic-statistical robust framework for Chinese AI-generated text detection", NeurIPS 2026, under review (student first author / co-first author). Contributions: LPP statistical features, MGAB held-out protocol, cross-dataset OOD design. Holdout n=33,739: Acc 98.75%, AUC 0.9992, MCC 0.9746; QA n=3,516: AUC 0.950.

SKILLS & TOOLS

Programming & data	Python (pandas / NumPy, backtesting and factor evaluation), SQL, Excel advanced functions and pivot tables, FastAPI, React / TypeScript
Platforms	Wind, Bloomberg, iFinD, WorldQuant BRAIN, Milvus, CLIP
Methods	PIT alignment, entity-mapping validity, coverage and denominator design, missing-value taxonomy, factor validation, cost and capacity estimation

EDUCATION

Beijing University of Posts and Telecommunications (BUPT) — B.Eng. in Computer Science and Technology
Sep 2024 - Jun 2028 (expected)
Rank 47/452 (top 10%). Coursework: Data Structures 100, OOP 100, Mathematical Modelling 99.

- National-level competition awards: 2 · Provincial / ministerial: 8 · University-level: 4
- University second-class scholarship; Merit Student; 2026 campus top-10 psychological-support volunteer (nominee)

CERTIFICATIONS & LANGUAGES

CFA: Level 2
Languages: IELTS 6.5; CET-4 and CET-6 passed. Reads English filings, prospectuses, overseas research and English data dictionaries; verifies field definitions in Bloomberg English UI and vendor documentation.